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Dynamic Light Scattering from Dispersive Time-Varying Metasurfaces: A Causal Floquet–Bloch Approach

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***Abstract* – Dispersive time-varying metasurfaces promise unprecedented scattering phenomena, ranging from Floquet harmonic generation to parametric amplification. Here, we introduce a causal Floquet–Bloch formulation for the scattering analysis of metasurfaces with arbitrary dispersion under temporal modulation. The proposed approach enables efficient calculation of reflected and transmitted harmonics while respecting causality. As a proof of concept, we study a strongly dispersive WS_2 metasurface and demonstrate asymmetric harmonic scattering and signatures of parametric amplification under appropriate modulation conditions.**

I. Introduction

Time-modulated structures enable nonreciprocity, frequency conversion, and parametric amplification beyond static systems [1–3]. In Floquet-driven metasurfaces, temporal periodicity induces frequency harmonics that mediate controlled energy exchange between electromagnetic waves and external modulation sources, drawing considerable attention to photonic time crystals and reconfigurable metasurfaces with programmable responses [4–5].

Modeling dispersive time-modulated metasurfaces presents fundamental challenges, as temporal modulation couples multiple frequency harmonics while material dispersion introduces resonant frequency dependence that must be treated consistently across all harmonics. This interplay becomes particularly demanding in realistic multilayer metasurfaces, where strong resonances and structural complexity coexist. Although dispersion and time modulation have been studied in canonical geometries [6–7], practical metasurface platforms require a unified and causal framework capable of treating arbitrary geometries and strong dispersion within a consistent formulation.

In this work, we develop a causal Floquet–Bloch formulation for the rigorous analysis of dispersive time-varying metasurfaces. The proposed framework incorporates arbitrary material dispersion within a harmonic coupling scheme while preserving physical causality and enabling both scattering and band-structure calculations. As a proof of concept, we apply the method to a strongly dispersive WS_2 metasurface and demonstrate asymmetric harmonic scattering together with signatures of parametric amplification under suitable modulation conditions.

II. Theoretical Platform

We consider a one-dimensional (1D) metasurface periodic along x with period Λ and uniform along y. The material properties are also periodic in time with period $T = 2\pi/\Omega$. Owing to the 1D geometry, the electromagnetic fields decouple into transverse electric (TE) and transverse magnetic (TM) subsystems. For brevity, we present the formulation for TE polarization; the TM case follows analogously.

In a dispersive time-varying medium, the polarization density is described by the causal constitutive relation

$$\mathbf{P}(x,z,t) = \varepsilon_0 \int_{-\infty}^{+\infty} \chi_e(x,t,\alpha)\, \mathbf{E}(x,z,t-\alpha)\, d\alpha, \tag{1}$$

where α denotes the delay variable. Under spatiotemporal periodicity, the electric susceptibility admits a Fourier expansion

$$\chi_e(x,t,\alpha) = \sum_{m,n} \chi_{mn}(\alpha)\, e^{imKx} e^{-in\Omega t}, \tag{2}$$

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with $K = 2\pi/\Lambda$. The harmonic coefficients $\chi_{mn}(\alpha)$ inherit the causality condition $\chi_{mn}(\alpha) = 0$ for $\alpha < 0$.

The electric field is expanded using the Floquet–Bloch form

$$E_y(x,z,t) = e^{ik_x x} e^{-i\omega t} \sum_{m,n} E_{mn}(z)\, e^{imKx} e^{-in\Omega t}, \tag{3}$$

where $K = 2\pi/\Lambda$. Substitution of this expansion, together with the Fourier series of the susceptibility, into Maxwell's equations yields a coupled eigenvalue problem along the propagation direction z. The resulting eigenvalue equation determines the allowed Floquet–Bloch propagation constants β_q and their corresponding modal profiles.

Accordingly, within each layer the electric field can be expanded as a superposition of eigenmodes,

$$E_y(x,z,t) = e^{ik_x x} e^{-i\omega t} \sum_{q} A_q\, e^{i\beta_q z} \sum_{m,n} \tilde{E}^{(q)}_{mn} e^{imKx} e^{-in\Omega t}, \tag{4}$$

where β_q denotes the Floquet–Bloch wavevector of mode q, $\tilde{E}^{(q)}_{mn}$ are the corresponding harmonic amplitudes, and A_q are excitation coefficients determined from boundary conditions.

At each interface, continuity of the tangential fields imposes

$$E_y^{(1)} = E_y^{(2)}, \qquad H_x^{(1)} = H_x^{(2)}, \tag{5}$$

for the full set of Floquet harmonics. These relations can be assembled into a scattering matrix formalism that relates incoming and outgoing harmonic amplitudes as $\mathbf{b} = \mathbf{S}\mathbf{a}$, where $\mathbf{a}$ and $\mathbf{b}$ denote the vectors of incident and scattered Floquet orders, respectively [8]. The S-matrix enables rigorous evaluation of reflected and transmitted powers via the time-averaged Poynting flux of each Floquet harmonic. Moreover, the same eigenvalue formulation provides direct access to the complex band structure of photonic time crystals based on periodically modulated metasurface slabs.

III. Results and Discussion

To showcase the functionality of the proposed platform, we investigate the scattering of light from a time-varying metasurface based on a monolayer WS_2 grating, as depicted in Fig. 1(a,b). The structure consists of periodically patterned WS_2 strips placed on a SiO_2 spacer backed by a silver mirror. The optical response of WS_2 is dominated by its strong A-exciton resonance in the visible regime, which is given as

$$\varepsilon_r(\omega,t) = \varepsilon_\infty + \frac{f(t)\,\omega_p^2}{\omega_0^2 - i\gamma_0\omega - \omega^2}. \tag{6}$$

where $\varepsilon_\infty = 18$ represents the high frequency permittivity, the excitonic resonance energy is $\hbar\omega_0 \approx 2.02$ eV, the damping rate is denoted as $\hbar\gamma_0 \approx 0.029$ eV, and the plasma frequency corresponds to $\hbar\omega_p \approx 1.38$ eV. Temporal modulation is introduced through the oscillator strength, as $f(t) = f_0\,(1 + \delta\cos(\Omega t))$ with $f_0 = 1$. This modulation directly alters the excitonic coupling strength while preserving the intrinsic resonance frequency and damping.

The structure is excited by a y-polarized, normally incident plane wave at $\lambda_0 = 617$ nm, corresponding closely to the excitonic resonance of WS_2. We first consider a slow modulation regime with $\Omega = \gamma_0/10$ and modulation depth $\delta = 0.1$. In this case, temporal modulation primarily induces frequency conversion, leading to the appearance of symmetric Floquet harmonics while preserving the overall dissipative character of the resonance. Next, we investigate the parametric regime by choosing $\Omega = 2\omega_0$ with $\delta = 0.1$. This condition enables parametric amplification through resonant coupling between harmonics. As shown in Fig. 1(d), energy transfer to the first-order harmonic is enhanced near the excitonic resonance, as a direct consequence of degeneracy between different harmonics. To further illustrate the impact of modulation strength, we increase the modulation depth to $\delta = 0.3$ while keeping $\Omega = 2\omega_0$, as shown in Fig. 1(e). The first-order harmonic amplitude increases significantly, indicating stronger parametric interaction.

However, due to the intrinsically lossy nature of WS_2, characterized by its relatively large damping rate γ_0, the signatures of parametric amplification are partially masked by material absorption. The dissipative excitonic response competes with modulation-induced energy transfer, limiting the observable net gain. Nevertheless, the clear growth of the first-order harmonic with increasing modulation depth confirms that the proposed dispersive Floquet–Bloch framework rigorously captures parametric effects in realistic, lossy excitonic metasurfaces.

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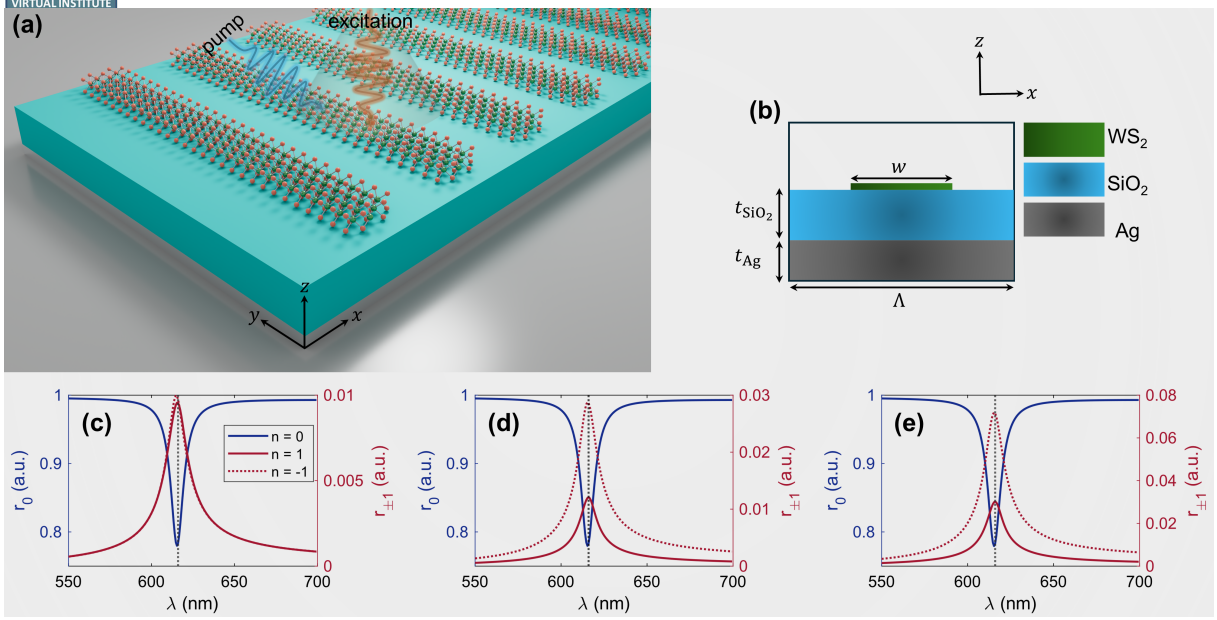


Fig. 1: (a) Schematic of the time-modulated WS_2 metasurface excited by an incident plane wave along the y-axis and driven by an external pump. (b) Cross section of the structure consisting of patterned WS_2 strips, placed on a SiO_2 spacer and backed by an Ag mirror ($\Lambda = 360$ nm, $w = 260$ nm, $t_{SiO_2} = 106$ nm, and $t_{Ag} = 100$ nm). (c–e) Reflected zeroth- and first-order Floquet harmonics for increasing modulation strength. Slow modulation ($\Omega = \gamma_0/10$, $\delta = 0.1$) induces frequency conversion, while parametric modulation ($\Omega = 2\omega_0$) with $\delta = 0.1$ and $\delta = 0.3$ enhances first-order harmonics near the excitonic resonance ($\lambda_0 = 617\,\mathrm{nm}$).

IV. Conclusion

We introduced a causal Floquet–Bloch framework for analyzing dispersive time-varying metasurfaces. The formulation rigorously incorporates material dispersion within a harmonic coupling scheme and enables both scattering and band-structure analysis. Application to a WS_2 metasurface demonstrates asymmetric harmonic generation and signatures of parametric amplification, highlighting the capability of the proposed platform for modeling realistic time-modulated photonic structures.

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